

Mohammad Mahmudul Hasan

Biomedical signal processing | Team Lead, AI Engineer

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RESEARCH PROFILE

Electrical engineering graduate and AI engineering team lead with undergraduate EEG research and a co-authored publication on explainable retinal OCT classification. Seeking graduate research in biomedical signal processing, interpretable learning and reliable physiological-state estimation.

EDUCATION

Islamic University of Technology, Bangladesh | BSc in Electrical and Electronic Engineering, completed 22 March 2021.

CGPA 3.76/4.00; first class with honours. Project and Thesis: A+; Digital Signal Processing: A; Artificial Neural Networks and Fuzzy Logic: A+.

PROFESSIONAL EXPERIENCE

HawalIT Limited — Team Lead, AI Engineer | with the company since 2021; team-lead role since [ADD month and year of promotion].

Currently lead a six-member AI engineering team.

- Worked with imagery and point clouds in production settings where data preparation, alignment and output quality influence downstream interpretation.
- Currently lead a six-member AI engineering team, bringing practical engineering coordination to a proposed return to biomedical research.

Earlier titles, exact dates and individual project ownership: [ADD verified career chronology].

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, Mohammad Mahmudul Hasan, Abrar Islam, and Md. Golam Rabiul Alam. “Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI.” 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2021. DOI: [10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

Individual contribution: [ADD precisely what you did]. The author preprint is the same work.

SELECTED RESEARCH AND TECHNICAL WORK

EEG ANALYSIS OF IMPAIRED CONSCIOUSNESS | Undergraduate thesis

Conducted undergraduate EEG signal-analysis research related to impaired consciousness and coma. Sleep-state analysis is an additional interest described in my current research direction.

Research question: Which EEG features generalize across people, and how should an uncertain brain-state estimate be represented?

EXPLAINABLE RETINAL OCT CLASSIFICATION | Peer-reviewed conference publication

Co-authored Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI. The study examines a compact CNN and LIME-based explanations.

Research question: How stable are explanations under image perturbations, and how should they be evaluated alongside classification and calibration?

MACHINE-FAULT ANALYSIS FROM VIBRATION SIGNALS | Earlier academic project

Worked with vibration time series and MATLAB-based signal feature extraction for a machine-fault detection project.

Research question: Which signal representations remain reliable when acquisition conditions change?

RESEARCH DIRECTION

How can physiological-state models communicate uncertainty and retain performance across people and recording conditions? Start with subject-independent sleep staging, compare spectral-feature and neural baselines, and measure calibration and explanation stability. A later extension to disorders of consciousness would require separate data, appropriate labels and clinical collaboration.

TECHNICAL FOUNDATION

Python; MATLAB; signal analysis; machine learning; image classification; data preparation. Specific biomedical libraries and current framework proficiency: [ADD verified tools].

ADDITIONAL CONTEXT

Independent AI/ML study during the final semesters of the EEE degree and participation in AI competitions. Historical Coursera credentials and competition placement should be added with year and verification links.

Language: Bangla (native); English-medium instruction certificate available. IELTS pending; intended overall score 7.0 is not an achieved score.

References: [ADD academic and professional referees as required by the program].